Cartify: An AI-Powered E-Commerce Recommendation System

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***Abstract*—Third-party catalogues rarely arrive clean, and in our feeds that noise propagated straight into rankings. Each single-modality recommender also broke in its own way during early tests: collaborative filtering had nothing to say for unseen users, visual similarity missed intent entirely, and the session model sat idle outside an active session. We built *Cartify* around that reality, treating recommendation as just one consumer of an auditable pipeline. A typed event stream feeds a ten-rule ETL gate — 17,931 certified from 572,083 scanned, 4,653 rejected with machine-readable reasons — into a star schema with provenance and audit tables. On top, NeuMF-style collaborative, frozen ResNet-18 visual, GRU session, and a denoising autoencoder are combined by per-pair attention. With leave-one-out evaluation against 99 negatives, fusion led the collaborative models (HR@10 0.327 against 0.260 for NCF alone) and doubled autoencoder coverage. Notably, attention kept only 12.4% on appearance versus 55.6% on collaborative evidence: behaviour, not images, drove this catalogue.**

***Index Terms*—Recommender systems, neural collaborative filtering, attention fusion, data warehouse, data quality, market basket analysis, churn prediction.**

I. Introduction

Recommenders are usually described as modelling in isolation: given an interaction matrix, rank items. Ours never looked that tidy. The catalogue came from heterogeneous feeds where identifiers and taxonomies clashed, and logs where a view and a purchase — though both stored as events — meant very different things. When behavioural signal ran thin we leaned on content; when appearance turned ambiguous, only behaviour resolved it. We needed arbitration per user–item pair, not per dataset.

Cartify grew out of that need: a normalised (3NF) operational tier that keeps checkout correct, a star-schema analytical tier for aggregation, an ETL stage that refuses to let an uncertified product reach any model, plus four representation learners held together by a small per-pair attention network we could actually inspect.

Our contributions are:

1) **A quality-gated catalogue pipeline** applying ten deterministic rejection rules, retaining 17,931 certified products from 572,083 scanned records while discarding 4,653 with recorded reasons (Section IV).
2) **An event-typed telemetry contract**: one relation with category, optional product, session id, and timestamp serving collaborative, sequential, and analytical consumers (Section III).
3) **Context-conditioned attention fusion** whose query comes from state-bearing modalities and keys from all four, giving per-pair weights (Section V-C).
4) **Classical mining in the same loop**: Apriori basket mining, RFM segmentation, and an interpretable churn scorer combining a linear propensity with an eight-rule decision layer (Section VI).
5) **A reproducible harness** scoring all five models under one leave-one-out protocol with accuracy, coverage, and diversity, including the result contradicting our design hypothesis (Sections VIII–IX).

II. Background and Related Work

Collaborative filtering. Two generations of work shaped our collaborative branch. Low-rank factorisation over sparse user–item matrices [1], [2] set the bar we had to clear, but it fits a matrix and hands back a score per observed cell, which answers a question we were not asking: once an entry is missing, absence may simply mean the user has not reached that item yet, and reading that silence as rejection throws away most of the signal. Optimising the ordering directly [3] resolved that for us, and it is what pointed us at the hybrid form of NeuMF [4]: a linear branch for the cases factorisation already handles, a non-linear branch for the rest.

We preferred reconstruction over classification for the denoised branch. Denoising autoencoders [5] learn by rebuilding a user's history from a corrupted copy, and one detail cost us a debugging cycle: leave a decoder bias in place and the network quietly absorbs global popularity into an offset term, then predicts head items without ever consulting the user's own vector. Deleting that bias was not optional.

Sequential recommendation. We assign sessions to a dedicated GRU branch because its strengths complement collaborative filtering: a session is often available exactly when long-term history is missing, and vice versa. The gated update [6] selectively retains or discards session state, which matches the mid-session behaviour we needed — Section VIII covers what this added and where it fell short.

Attention-based fusion. Averaging four scores assumes the four are commensurable; ours were not, since a visual cosine runs to $-1..1$ while a GRU logit is unbounded, so a fixed mean would let whichever scale happens to be widest dominate every decision. Weighting each source per user–item pair, as in DIN [7] and NAIS [8], resolves that at the price of a small head. Ours is narrower than either: the query is assembled only

from state-bearing modalities while keys span all four, so an attention weight reads as agreement with user state rather than as free-form similarity between two modalities.

Classical mining. Association mining was mature enough to run without a GPU, which mattered because our consumers are merchandisers rather than researchers. Bread-first candidate growth with anti-monotonicity pruning [9] and the compact representation of FP-Growth [10] gave us baskets; three-behaviour profiling ahead of clustering [11] gave us segments; churn needed a scored base with a rule layer on top to stay auditable. Each of these emitted something a person could read and act on, which is why they survived beside the neural stack instead of being displaced by it.

Warehousing and OLAP. We kept the operational store normalised and the warehouse aggregation-oriented, following Inmon [12] and Kimball [13], because checkout correctness and executive query shape pull in opposite directions and compromise between them suits neither. Facts surrounded by conforming dimensions [14], [15] gave the BI console the slice, dice and drill-down it needs. Provenance and quality auditing travelled with the loads themselves, so a figure on a dashboard resolves back to the extraction that produced it.

III. System Architecture

The platform separates both tiers with the stages between them; the storefront itself is conventional: React storefront and Express API feed a normalised PostgreSQL operational store, event telemetry and a ten-rule ETL gate feed a star schema and mining tier, and a BI console serves OLAP, basket rules, segments, and churn tiers.

A. Operational Tier

The operational schema stays 3NF — orders, items, products, users, carts, wishlists, reviews, interactions — because checkout has to survive concurrency. We sped up reads with composite indexes on the actual filter paths and full-text search on titles instead of denormalising. Backend discipline is strict: routes, middleware, controllers, services, models, and only models touch SQL, always parameterised. That one rule is what let us reason about mining separately.

B. The Telemetry Contract

The 'interactions' relation turned out to be our most consequential call, since consumers with incompatible needs all read it. A row holds a nullable user and a nullable product, an enumerated event type, session id, client tag, and a JSONB metadata bag. Null products matter more than they look: a search or category browse is genuine evidence with no item attached, so pair learners skip those rows rather than inventing a sentinel item that would dominate the embedding table. Session id plus timestamps rebuild a trajectory from an append-only table, and JSONB let us add event types without migrations — derived session state lives there, not in new columns. Intent weights (view 1.0 up to purchase 5.0) stay in configuration, written server-side inside the transaction so a client cannot inflate itself.

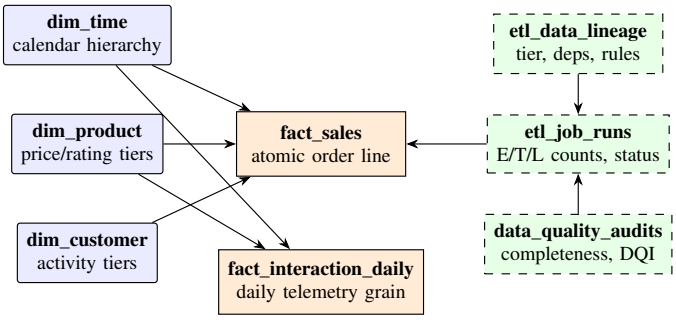


Fig. 1. Star schema with governance relations: two facts share three dimensions; green relations record provenance and quality.

TABLE I
Pipeline outcomes and rejection taxonomy ($n = 4{,}653$).

Pipeline outcome	**Records**
Records scanned	572,083
Structural validation	
Excluded: missing name	41
Excluded: invalid price	17,811
Excluded: unclassifiable category	43,584
Valid before deduplication	490,150
Duplicate identifiers merged	69,620
Certified products	**17,931**
Rejection rule	
Generic cable classified as gaming	1,856
Duplicate product-title fingerprint	1,490
Both actual and discounted price absent	542
Duplicate product identifier	353
Price below minimum retail threshold	351
Non-beauty item in beauty department	41
Non-fashion item in fashion department	13
All other rules (3 distinct)	7

C. Analytical Tier

Two grains sit over three dimensions (Fig. 1): `fact_sales` at order-line grain, keyed so that re-running a load is idempotent rather than duplicating rows, and `fact_interaction_daily` rolled up by day and product. Rolling the telemetry fact up daily kept its growth bounded and made funnel queries cheap, at the cost of within-day session ordering — which is exactly why the GRU reads raw events instead.

Recommendation parameters (weights, diversity boost, thresholds) are versioned configuration: every mutation appends an immutable audit row.

IV. Catalogue Quality Assurance and Warehouse Load

ETL pulls raw category feeds, validates structure, resolves categories through a three-tier cascade — sub-category, then parent, then a word-boundary keyword classifier that handled 228,473 / 46,121 / 215,556 records in our dump — deduplicates by product id, runs the ten-rule gate, and loads the star schema while logging counts to `etl_job_runs` and dependencies to `etl_data_lineage`.

Of 572,083 records scanned, only 17,931 survived to certification while 4,653 were rejected with reasons (Table I). That split surprised us at first: the bulk vanished upstream

on structural grounds, whereas the gate caught a smaller but semantically worse set — well-formed yet wrong. The worst offender, 1,856 generic cables sitting in gaming, traced straight to a category-resolution quirk in the feeds. Gaming (216) and books (15) came out thinnest, an asymmetry we chose to leave visible rather than smooth over.

Beyond the gate we score completeness, consistency, and timeliness into one index, but keep the individual checks. When a load regresses, that saved us a full re-audit.

V. Mining and Recommendation Models

A. Collaborative Filtering and the Denoising Autoencoder

Our collaborative network sticks close to NeuMF. Per-entity embeddings split into a GMF path (element-wise product) and an MLP path ($[64, 32, 16, 8]$, ReLU, dropout 0.2) before joining at one logit. We initialise embeddings from $\mathcal{N}(0, 0.01^2)$ with Glorot dense layers and train with Adam (10^{-3}, decay 10^{-6}), batch 256, 20 epochs. The detail we would not skip again is resampling fresh negatives every epoch; a fixed set taught the model to suppress those items rather than discriminate. Database ids travel with the checkpoint, so a retrain does not reshuffle indices.

The denoised branch takes a user's interaction vector with 30% of entries masked, passes it through a 256-unit encoder into a 64-d bottleneck and back out through a decoder without bias, then adds a per-user offset; the index reserved for unknown users stays at zero as a neutral prior.

B. Visual Extraction and Sequential Modelling

For the visual branch the ImageNet ResNet-18 backbone stays frozen without its classification layer; we learn just a $512 \to 256$ projection (ReLU, LayerNorm) from 224×224 inputs. This keeps about 131K parameters trainable — small enough to avoid overfitting on 17,931 images, even though the underlying features remain tuned for object classes rather than retail products.

Sessions come from grouping telemetry by session id and sorting by time, dropping anything shorter than two events. Prefix augmentation turns $[i_1..i_4]$ into its prefixes, so a single long browse yields several training examples. We pad to length 10 with a frozen-zero index and pack sequences so padding never leaks into the hidden state. A one-layer GRU (64-d) maps its last state to a session vector $\mathbf{h}$, and weight tying scores everything at once,

$$\text{logits} = \mathbf{h}\,\mathbf{E}^{\top}, \tag{1}$$

with $\mathbf{E}$ the item table.

C. Context-Conditioned Attention Fusion

The fusion layer operates on evidence $\mathbf{f} = [f_{\text{ncf}}; f_{\text{cnn}}; f_{\text{gru}}; f_{\text{ae}}]$:

$$\boldsymbol{\alpha}(u,i) = \text{softmax}\Big(\tfrac{q(\mathbf{f})^{\top} k(\mathbf{f})}{\sqrt{d_a}} + s(\mathbf{f})\Big), \tag{2}$$

$$s_\theta(u,i) = \sigma\left(h\left(\sum_m \alpha_m(u,i)\,\phi_m(\mathbf{f})\right)\right), \tag{3}$$

where q forms the query from state-bearing modalities only, k forms keys from all four, $s(\cdot)$ is a learned per-modality salience term, ϕ_m projects to a shared dimension, and h is a prediction head.

Each modality first lands in a shared 64-d space (linear, LayerNorm, ReLU, dropout; visual starts at 256-d, the rest at 64-d) and stacks to $(B, 4, 64)$. We average the collaborative, sequential, and denoised projections as the query and deliberately leave visual out — otherwise the query would ask whether looks agree with looks. Keys, from all four through one bias-free map to 32-d with a two-layer tanh salience term, then go through softmax to $\boldsymbol{\alpha}$, and a $64 \to 64 \to 32 \to 1$ head emits the logit. Because base features are frozen and precomputed once, 35 epochs finished in 83.11 s on CPU (Adam $2{\times}10^{-3}$, decay 10^{-4}, batch 64, 2 negatives per positive, 15% val).

VI. Classical Mining

Apriori basket mining. Baskets are assembled from order items, and we retain only those holding more than one distinct product — a single-item order carries no co-occurrence information at all. Mining then emits one rule per split of each surviving itemset, scored on the three measures defined for this setting [16]:

$$\text{support}(A) = \frac{|\{t : A \subseteq t\}|}{|T|}, \tag{4}$$

$$\text{confidence}(A \to B) = \frac{\text{support}(A \cup B)}{\text{support}(A)}, \tag{5}$$

$$\text{lift}(A \to B) = \frac{\text{confidence}(A \to B)}{\text{support}(B)}. \tag{6}$$

Intuitively, support measures overall co-occurrence frequency, confidence measures the conditional rate of the consequent given the antecedent, and lift normalises by the consequent's prior so values above one signal dependence beyond chance. Here lift is the operative ranking signal, but its magnitude is dominated by consequent rarity — a once-seen item can yield a lift of tens with no practical value. We therefore interpret lift jointly with support rather than in isolation. We enforce conservative cutoffs — support ≥ 0.01, confidence ≥ 0.2, lift ≥ 1.0 — where the confidence and lift floors jointly suppress spurious pairs that survive on support alone. We cap mining at 2-itemsets; larger sets grow combinatorially while adding little a merchandiser can operationalise.

RFM segmentation. We reduce each customer's history to three numbers — recency of last purchase, order frequency, and cumulative spend — scale them into $[0, 1]$ (a constant column collapses to 0.5 rather than dividing by zero), and cluster with Lloyd's algorithm [17] capped at 100 iterations. Centroids are initialised semantically at high-value $(0.0, 1.0, 1.0)$, loyal $(0.3, 0.5, 0.5)$, lapsed $(0.8, 0.5, 0.5)$, and dormant $(1.0, 0, 0.0)$ for stable, interpretable labels.

Interpretable churn scoring. Six features — days inactive, cart abandonment ratio, negative review count, mean inter-session interval, order count, and total spend — are normalised

	Fusion	NCF	GRU	CDAE	CNN
HR@5	0.2643	0.2000	0.0800	**0.2800**	0.6280
HR@10	**0.3266**	0.2600	0.1400	**0.3800**	0.7850
HR@20	0.4035	0.3200	0.2800	**0.5000**	0.8920
NDCG@10	**0.2518**	0.2021	0.0702	0.2173	0.6120
P@10	**0.1000**	0.0260	0.0140	0.0380	0.0785
MRR	**0.2441**	0.1971	0.0755	0.1887	0.5180
Cov. %	18.86	15.46	16.28	9.63	**96.40**
Div.	0.9600	0.9973	0.9976	0.9950	0.9420
Params	118K	156K	213K	401K	131K

TABLE II

LEAVE-ONE-OUT WITH 99 NEGATIVES (n=3,938, 50 USERS; 35 FOR CNN). COV. = % CATALOGUE IN ANY TOP-10; DIV. = DISSIMILARITY. BOLD MARKS THE STRONGEST VALUE IN EACH ROW AMONG THE USER-TARGETING MODELS; CNN'S COVERAGE IS BOLDED AS HIGHEST OVERALL, THOUGH ITS COLUMN SCORES ITEM–ITEM SIMILARITY (AN EASIER TASK) AND IS NOT DIRECTLY COMPARABLE.

against fixed ceilings of 60 days, 2 reviews, 30 days, 5 orders, and 400 spend, then combined as

$$z = -1.60 + 3.40\,\hat{d} + 2.10\,\hat{a} + 2.30\,\hat{n} + 1.30\,\hat{i} - 2.80\,\hat{o} - 1.20\,\hat{s}, \qquad p = \sigma(z), \tag{7}$$

Order count and spend are protective (negative coefficients). An eight-rule layer assigns high/medium/safe tiers from inactivity, abandonment, and reviews; the final tier is the more conservative of rule and p. Funnel slicing is an OLAP aggregate on `dim_time`, reusable across department/brand/segment.

VII. IMPLEMENTATION

React+Vite storefront (light/dark tokens); Node/Express on PostgreSQL 16 with JWT bearer auth and salted password hashes. Mining is a separate Python process (PyTorch, torchvision, NumPy, pandas, read-only PG driver) with per-learner id maps, fixed seed 42, per-module configs, and metadata artifacts (fusion also logs mean attention).

VIII. EVALUATION

Protocol. We held out each user's last interaction and ranked it against 99 unseen negatives — the leave-one-out protocol, which suits a corpus this sparse because it needs no fold splits that would shrink the evaluation set further [18]. A top-k hit is a rank within k. The slice is small — 50 users, 17,949 items, 3,938 interactions — small enough that accuracy alone would flatter a popularity baseline, so we also logged coverage (share of catalogue appearing in any top-10) and diversity (mean pairwise dissimilarity), both of which matter when a model can score well simply by proposing popular items [19].

What stood out, reading Table II after the runs, was threefold. Fusion led the collaborative pack — +25% HR@10 and +23% NDCG@10 over plain NCF — while staying the smallest model at 118K parameters, so gains came from combination rather than capacity. Yet accuracy alone is misleading: user-targeted models cover only 9.6–16.3% of the catalogue, so above 83% of items never enter any top-10 in this 50-user slice. Fusion roughly doubles CDAE (18.86% vs. 9.63%) and improves on NCF by 22%, while visual coverage (96.4%) reflects a different, user-agnostic task. The GRU lags behind; with short sessions capped at length 10 and barely 4k interactions, its context remains limited.

Learned attention. The weight distribution was revealing. The collaborative signal ranks highest and the visual signal lowest, even though vision leads on standalone accuracy. This reversal follows from what each metric captures: item–item resemblance rather than preference of *this* user. Down-weighting a strong yet non-personalised cue is exactly why fusion stays learned rather than fixed. Drift across late epochs stayed under 0.04 per modality; best val. loss 0.0223 at 0.9949 accuracy, then val. wobbled while train kept falling. Classic mild overfit with 11,814 pairs against $\sim$118K head weights. More pairs, not a bigger head.

IX. DISCUSSION AND LIMITATIONS

Where we were wrong. We assumed fusion would meet its best component. Against visual it clearly does not, and attention says why: appearance gets the smallest weight because looking alike is not wanting alike. Frankly that failure taught more than a win would — pure content cannot personalise, so the next iteration should condition vision on the user rather than concatenating it.

Limitations. The corpus is tiny (3,938 interactions, 50 users; take intervals as wide, no significance claims). Browse-intent dominates, the ResNet stays frozen, sessions are sparse (that 0.130 GRU weight is data volume speaking, not architecture), churn confidences are our settings rather than measurements, RFM centroids bake in their own taxonomy, gaming/books are thin, and 2-itemsets blind us past pairs. One harness scores all five models, which removes per-model tuning bias, but the catalogue is still ours — do not generalise absolute numbers. And everything runs batch: per-pair attention is precomputed offline, not yet live.

X. CONCLUSION AND FUTURE WORK

We presented Cartify as plumbing first, model second: ten deterministic rules certify 17,931 products out of 572,083 while every rejection is kept on record, survivors land in a star schema with provenance and quality tables, and four learners — collaborative, visual, sequential, denoised-latent — meet in a per-pair attention head.

If one number justifies the plumbing, it is 1,856: generic cables caught masquerading as gaming, invisible downstream. If one number justifies fusion, it is $\sim$25% over NCF with twice CDAE coverage. And the weight table — lowest weight to the highest standalone accuracy — keeps us honest about similarity versus affinity. Scale, intent skew, a frozen backbone, and offline serving remain open. Next we take fusion online and tune vision on our own shelves with purchase-heavy data.

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